Exercise J-M

1. If $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$, $x \in R$, then : **[JEE(Main)Jan.2023]**

यदि $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$, $x \in R$ है, तब : **[JEE(Main)Jan.2023]**

(A) $2f(0)-f(1)+f(3)=f(2)$ (B) $f(1)+f(2)+f(3)=f(0)$

(C) $3f(1)+f(2)=f(3)$ (D) $f(3)-f(2)=f(1)$

Ans. A

Sol. $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$

$f'(x)=3x^2-2xf'(1)+f''(2)$

$f''(x)=6x-2f'(1)$

$f'''(x)=6 \Rightarrow f'''(3)=6$

$\text{from (3)} \rightarrow f''(2)=12-2f'(1)$

$\text{from(2)} \rightarrow f'(1)=3(1)^2-2f'(1)+f''(2)$

$\Rightarrow f''(2)=3f'(1)-3$

$\Rightarrow 12-2f'(1)=3f'(1)-3 \Rightarrow f'(1)=3$

$f''(2)=12-6=6$

$f(x)=x^3-3x^2+6x-6, \quad f(0)=-6$

$f(1)=-2, \quad f(2)=2, \quad f(3)=12$

2. Let $y(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y'-y''$ at $x=-1$ is equal to : **[JEE(Main)Jan.2023]**

माना $y(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$ है। तो $x=-1$ पर $y'-y''$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 464 (B) 944 (C) 496 (D) 976

Ans. C

Sol. $y=(1+x)\left(1+x^2\right)\left(1+x^4\right)\left(1+x^8\right)\left(1+x^{16}\right)$

$\Rightarrow y=\dfrac{(1-x)(1+x)\left(1+x^2\right)\left(1+x^4\right)\left(1+x^8\right)\left(1+x^{16}\right)}{(1-x)}$

$\Rightarrow y=\dfrac{1-x^{32}}{(1-x)}$

$\Rightarrow (1-x)y=1-x^{32}$

$\Rightarrow (1-x)y'-y=-32x^{31}$

$\Rightarrow (1-x)y''-2y'=-32.31x^{30}$

Put $x=-1$

$\Rightarrow y' - y'' = 496$

3. Let $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right) - 2(1 - \sin^2 2\theta)$ and

$S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$. If $4\beta = \sum_{\theta \in S} \theta$, then $f(\beta)$ is equal to : **[JEE(Main)Jan.2023]**

माना $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right) - 2(1 - \sin^2 2\theta)$ एवं

$S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$ है। यदि $4\beta = \sum_{\theta \in S} \theta$, है तब $f(\beta)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{11}{8}$ (B) $\frac{3}{2}$ (C) $\frac{9}{8}$ (D) $\frac{5}{4}$

Ans. D

Sol. $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3x + \theta)\right) - 2\left(1 - \sin^2 2\theta\right)$

$S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$

$\Rightarrow f(\theta) = 3(\cos^4\theta + \sin^4\theta) - 2\cos^2 2\theta$

$\Rightarrow f(\theta) = 3\left(1 - \frac{1}{2}\sin^2 2\theta\right) - 2\cos^2 2\theta$

$\Rightarrow f(\theta) = 3 - \frac{3}{2}\sin^2 2\theta - 2\cos^2 2\theta$

$= \frac{3}{2} - \frac{1}{2}\cos^2 2\theta = \frac{3}{2} - \frac{1}{2}\left(\frac{1 + \cos 4\theta}{2}\right)$

$f(\theta) = \frac{5}{4} - \frac{\cos 4\theta}{4}$

$f'(\theta) = \sin 4\theta$

$\Rightarrow f'(\theta) = \sin 4\theta = -\frac{\sqrt{3}}{2}$

$\Rightarrow 4\theta = n\pi + (-1)^n \frac{\pi}{3}$

$\Rightarrow \theta = \frac{n\pi}{4} + (-1)^n \frac{\pi}{12}$

$$\Rightarrow \theta = \frac{\pi}{12}, \left(\frac{\pi}{4} - \frac{\pi}{12}\right), \left(\frac{\pi}{2} + \frac{\pi}{12}\right), \left(\frac{3\pi}{4} - \frac{\pi}{12}\right)$$

$$\Rightarrow 4\beta = \frac{\pi}{4} + \frac{\pi}{2} + \frac{3\pi}{4} = \frac{3\pi}{2}$$

$$\Rightarrow \beta = \frac{3\pi}{8} \Rightarrow f(\beta) = \frac{5}{4} - \frac{\cos\frac{3\pi}{2}}{4} = \frac{5}{4}$$

4. Let $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$. Then, at $x = 1$: **[JEE(Main)Jan.2023]**

माना $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$ है। तो $x = 1$ पर **[JEE(Main)Jan.2023]**

(A) $\sqrt{2}y' - 3\pi^2 y = 0$ (B) $y' + 3\pi^2 y = 0$

(C) $2y' + 3\pi^2 y = 0$ (D) $2y' + \sqrt{3}\pi^2 y = 0$

Ans. C

Sol. $Y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{3/2}\right)\right)\right)$

$$\frac{dy}{dx} = 3\sin^2\left(\frac{\pi}{3}\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)\right)^{3/2}\right) \times \frac{\pi}{3}\left(-\sin\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x + 1\right)\right)\right)^{3/2}$$

$$\left(\cos\left(\frac{\pi}{3}\cos\left(\frac{\pi}{3\sqrt{2}}(-4x^3 + 5x^2 + 1)^{3/2}\right) \times \frac{\pi}{3\sqrt{2}}(3/2(-4x^3 + 5x^2 + 1)^{1/2})(-12x^2 + 10x\right)\right)$$

at $x = 1$

$$y = f(1) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}.2\sqrt{2}\right)\right)\right)$$

$$= \sin^3\left(\frac{\pi}{3}(-1/2)\right)$$

$$= \sin^3\left(-\frac{\pi}{6}\right) = \left(-\frac{1}{2}\right)^3 = -\frac{1}{8}$$

$$\left(\frac{dy}{dx}\right)_{x=1} = 3\left(-\frac{1}{2}\right)^2 \cdot \left(-\frac{\pi}{3}\right)\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) \cdot \frac{\pi}{3\sqrt{2}}\left(\frac{3}{2}\sqrt{2}\right)(-2)$$

$$= \frac{3}{4}\cdot\frac{\pi}{3}\cdot\frac{3}{4} = \frac{3\pi^2}{16}$$

Now (3) $$2y' + 3\pi^2 y = 2\left(\frac{3\pi^2}{16}\right) + 3\pi^2\left(-\frac{1}{8}\right) = \frac{3\pi^2}{8} - \frac{3\pi^2}{8} = 0$$

5. If $y(x) = x^x, x > 0$, then y"(2) – 2y'(2) is equal to : **[JEE(Main)Jan.2023]**

यदि $y(x) = x^x, x > 0$ है, तो y"(2) – 2y'(2) बराबर है : **[JEE(Main)Jan.2023]**

(A) $4(\log_e 2)^2 + 2$ (B) $4\log_e 2 + 2$

(C) $8\log_e 2 - 2$ (D) $4(\log_e 2)^2 - 2$

Ans. D

Sol. $y' = x^x$

$y' = x^x + (1 + \ln x)$

$$y'' = x^x(1+\ln x)^2 + x^x.\frac{1}{x}$$

$y''(2) = 4(1 + \ln 2)^2 + 2$

$y'(2) = 4(1 + \ln 2)$

$y''(2) - 2y'(2) = 4(1 + \ln 2)^2 + 2 - 8(1 + \ln 2)$

$= 4(1 + \ln 2)[1 + \ln 2 - 2] + 2$

$= 4(\ln 2)^2 - 1) + 2$

$= 4(\ln 2)^2 - 2$

6. If $f(x) = x^2 + g'(1)x + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$, then the value of f(4) – g(4) is equal to _____. **[JEE(Main)Jan.2023]**

यदि $f(x) = x^2 + g'(1)x + g''(2)$ एवं $g(x) = f(1)x^2 + xf'(x) + f''(x)$ है, तो f(4) – g(4) का मान है। **[JEE(Main)Jan.2023]**

Ans. 14

Sol. $f(x) = x^2 + xg'(1) + g''(2)$ ...(1)

$\therefore f'(x) = 2x + g'(1)$

$f''(x) = 2$

$\therefore g(x) = f(1)x^2 + x(2x + g'(1)) + 2 = f(1)x^2 + 2x^2 + xg'(1) + 2$

$g'(x) = 2f(1)x + 4x + g'(1)$ ...(2)

$g''(x) = 2(1) + 4$ ...(3)

from (2) put x = 1

$g'(1) + 2f(1) + 4 + g'(1)$

$\Rightarrow 2f(1) + 4 = 0$

$\Rightarrow f(1) = -2 \Rightarrow g''(2) = -4 + 4 = 0$ from (3)

from (1)

$f(1) = 1 + g'(1) + 0$

$\Rightarrow g'(1) = -2 - 1 = -3$

$\because f'(x) = 2x - 3$

$f(x) = x^2 - 3x + c$

put $x = 1, f(1) = -2 \quad \therefore c = 0$

$\therefore f(x) = x^2 - 3x$

$\therefore g(x) = -3x + 2$

$\Rightarrow f(4) - g(4) = 4 - (-10) = 14$

7. If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at $(2, 2)$ is equal to : **[JEE(Main)April,2023]**

यदि $2x^y + 3y^x = 20$ है, तब $(2, 2)$ पर $\frac{dy}{dx}$ का मान है : **[JEE(Main)April,2023]**

(A) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$ (B) $-\left(\frac{3+\log_e 4}{2+\log_e 8}\right)$ (C) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$ (D) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$

Ans. D

Sol. $2x^y + 3y^x = 20$

$$2x^y\left[\frac{y}{x} + (\ln x)y'\right] + 3y^x\left[\frac{xy'}{y} + \ln y\right] = 0$$

$$y' = \frac{-(12\ln 2 + 8)}{12 + 8\ln 2} = -\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$$

8. Let $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}, x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to : **[JEE(Main)April,2023]**

यदि $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}, x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$ है, तो $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{2}{9}$ (B) $\frac{-2}{3}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{-1}{3\sqrt{3}}$

Ans. A

Sol. $$f(x) = \sqrt{2}\frac{\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right) - 1}{\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x - \frac{1}{\sqrt{2}}\cos x\right)} = \frac{\sin(x + \pi/4) - 1}{\sin(x - \pi/4)}$$

$$f'(x)=\frac{\cos(x+\pi/4)\sin(x-\pi/4)-\cos(x-\pi/4)(\sin(x+\pi/4)-1)}{\sin^2(x-\pi/4)}$$

$$f'(x)=\frac{\cos(x-\pi/4)-1}{\sin^2(x-\pi/4)}=\frac{-(1-\cos(x-\pi/4))}{1-\cos^2(x-\pi/4)}$$

$$f'(x)=-\frac{1}{1+\cos(x-\pi/4)}$$

$$=f''(x)=-\frac{\sin(x-\pi/4)}{(1+\cos(x-\pi/4))^2}$$

$$f(7\pi/12)=-\frac{1}{\sqrt{3}}$$

$$f''(7\pi/12)=-\frac{2\sqrt{3}}{9}$$

$$f(7\pi/12)f''(7\pi/12)=\frac{2}{9}$$

9. For the differentiable function $f:\mathbb{R}-\{0\}\to\mathbb{R}$, let $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10$, then $\left|f(3)+f'\left(\frac{1}{4}\right)\right|$ is equal to:

[JEE(Main)April,2023]

अवकलनीय फलन $f:\mathbb{R}-\{0\}\to\mathbb{R}$ के लिए माना $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10$, है, तो $\left|f(3)+f'\left(\frac{1}{4}\right)\right|$ बराबर है :

[JEE(Main)April,2023]

(A) $\frac{33}{5}$ (B) 7 (C) $\frac{29}{5}$ (D) 13

Ans. D

Sol. $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10$ ...(i)

$$x\to\frac{1}{x}$$

$$3f\left(\frac{1}{x}\right)+2f(x)=x-10 \quad \text{...(ii)}$$

From (i) & (ii)

$$5f(x)=\frac{3}{x}-2x-10$$

Now, f(3) = –3

$$5f'(x)=\frac{-3}{x^2}-2$$

$$5f'\left(\frac{1}{4}\right) = -48 - 2$$

$$f'\left(\frac{1}{4}\right) = -10$$

$$\left|f(3) + f'\left(\frac{1}{4}\right)\right| = 13$$

10. Suppose $f(x) = \dfrac{\left(2^x + 2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2 - x + 1\right)}}{\left(7x^2 + 3x + 1\right)^3}$. Then the value of f'(0) is equal to :

[JEE(Main)Jan.2024]

माना $f(x) = \dfrac{\left(2^x + 2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2 - x + 1\right)}}{\left(7x^2 + 3x + 1\right)^3}$ है। तो f'(0) का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\dfrac{\pi}{2}$ (B) π (C) 0 (D) $\sqrt{\pi}$

Ans. D

Sol. Let f(x) = tan x g(x)

where $g(x) = \dfrac{\left(2^x + 2^{-x}\right)\sqrt{\tan^{-1}\left(x^2 - x + 1\right)}}{\left(7x^2 + 3x + 1\right)^3}$

$$\Rightarrow f'(x) = \sec^2 x g(x) + \tan x g'(x)$$

$$f'(0) = \sec^2(0) g(0) + \tan 0 g'(0)$$

$$= g(0)$$

$$g(0) = \frac{\left(2^0 + 2^{-0}\right)\sqrt{\tan^{-1}\left(0^2 - 0 + 1\right)}}{\left(7(0)^2 + 3(0) + 1\right)^3}$$

$$= 2\sqrt{\tan^{-1} 1} = 2\sqrt{\frac{\pi}{4}} = \sqrt{\pi}$$

11. Let $y = \log_e\left(\dfrac{1 - x^2}{1 + x^2}\right)$, $-1 < x < 1$. Then at $x = \dfrac{1}{2}$, the value of $225(y' - y'')$ is equal to :

[JEE(Main)Jan.2024]

माना $y = \log_e\left(\frac{1-x^2}{1+x^2}\right), -1 < x < 1$ है। तो $x = \frac{1}{2}$ पर $225(y' - y'')$ का मान है : **[JEE(Main)Jan.2024]**

(A) 736 (B) 746 (C) 732 (D) 742

Ans. A

Sol. $y = \log_e\left(\frac{1-x^2}{1+x^2}\right)$

$\frac{dy}{dx} = y' = \frac{-4x}{1-x^4}$

Again $\frac{d^2y}{dx^2} = y'' = \frac{-4\left(1+3x^4\right)}{\left(1-x^4\right)^2}$

Again $y' - y'' = \frac{-4x}{1-x^4} + \frac{4\left(1+3x^4\right)}{\left(1-x^4\right)^2}$

at $x = \frac{1}{2}$

$y' - y'' = \frac{736}{225}$

Thus $225\left(y' - y''\right) = 225 \times \frac{736}{225} = 736$

12. If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to : **[JEE(Main)Jan.2024]**

यदि $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ है, तो $\frac{1}{5}f'(0)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 6 (C) 0 (D) 1

Ans. C

Sol. $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$

$$f'(x)=\begin{vmatrix}8\cos^3 x(-\sin x) & 2\sin^4 x & 3+\sin^2 2x\\ 8\cos^3 x(-\sin x) & 2\sin^4 x & \sin^2 2x\\ 8\cos^3 x(-\sin x) & 3+2\sin^4 x & \sin^2 2x\end{vmatrix}+\begin{vmatrix}2\cos^4 x & 8\sin^3 x\cos x & 3+\sin^2 2x\\ 3+2\cos^4 x & 8\sin^3 x\cos x & \sin^2 2x\\ 2\cos^4 x & 8\sin^3 x\cos x & \sin^2 2x\end{vmatrix}+$$

$$\begin{vmatrix}2\cos^4 x & 2\sin^4 x & 4\sin 2x\cos 2x\\ 3+2\cos^4 x & 2\sin^4 x & 4\sin 2x\cos 2x\\ 2\cos^4 x & 3+2\sin^4 x & 4\sin 2x\cos 2x\end{vmatrix}$$

$$f'(0)=\begin{vmatrix}0&0&0\\0&0&0\\0&0&0\end{vmatrix}+\begin{vmatrix}2&0&0\\5&0&0\\2&0&0\end{vmatrix}+\begin{vmatrix}2&0&0\\5&0&0\\2&0&0\end{vmatrix}$$

$f(0)=0$

$\therefore \frac{1}{5}f'(0)=0$

13. If $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$ for all $x\in R$, then $2f(0)+f'(0)$ is equal to :

[JEE(Main)Jan.2024]

यदि सभी $x\in R$ के लिए $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$ हैं, तो $2f(0)+f'(0)$ बराबर है :

[JEE(Main)Jan.2024]

(A) 48 (B) 18 (C) 24 (D) 42

Ans. D

Sol. $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$

$$f'(x)=\begin{vmatrix}3x^2 & 4x & 3\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-x\end{vmatrix}+\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 6x & 2 & 3x^2\\ x^3-x & 4 & x^2-2\end{vmatrix}+\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ 3x^2-1 & 0 & 2x\end{vmatrix}$$

$$2f(0)+f'(0)=2\begin{vmatrix}0&1&1\\2&0&6\\0&4&-2\end{vmatrix}+\begin{vmatrix}0&0&3\\2&0&6\\0&4&-2\end{vmatrix}+\begin{vmatrix}0&1&1\\0&2&0\\0&4&-2\end{vmatrix}+\begin{vmatrix}0&1&1\\2&0&6\\-1&0&0\end{vmatrix}$$

$= 24 + 24 + 0 - 6$

$= 42$

14. Let g(x) be a linear function and $f(x)=\begin{cases} g(x) & , \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , \quad x > 0 \end{cases}$, is continuous at x = 0. If f′(1) = f(–1), then the value of g(3) is : **[JEE(Main)Jan.2024]**

माना g(x) एक रैखिक फलन है तथा $f(x)=\begin{cases} g(x) & , \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , \quad x > 0 \end{cases}$, x = 0 पर संतत है। यदि f′(1) = f(–1) है, तो g(3) का eku gS : **[JEE(Main)Jan.2024]**

(A) $\log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$ (B) $\log_e\left(\frac{4}{9}\right)-1$ (C) $\frac{1}{3}\log_e\left(\frac{4}{9}\right)+1$ (D) $\frac{1}{3}\log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$

Ans. A

Sol. Let g(x) = ax + b

f(x) is continuous at x = 0

$$\therefore \lim_{x\to 0^+}\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}=\left(\frac{1}{2}\right)^{\infty}=0$$

$$\lim_{x\to 0^-} g(x)=\lim_{x\to 0^-} ax+b=b$$

So, b = 0

Now, $f(x)=\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}$

$$\ell n f(x)=\frac{1}{x}\left[\ell n(1+x)-\ell n(2+x)\right]$$

$$\frac{1}{f(x)}\cdot f'(x)=\frac{x\left[\frac{1}{1+x}-\frac{1}{2+x}\right]-\ell n\left(\frac{1+x}{2+x}\right)}{x^2}$$

$$f'(1) = f(1)\left[\frac{\left(\frac{1}{2}-\frac{1}{3}\right)-\ell n\left(\frac{2}{3}\right)}{1}\right]$$

$$f'(1) = \frac{2}{3}\left[\frac{1}{6}-\ell n\frac{2}{3}\right] \quad ...(1)$$

Also $f(-1)\ a(-1)+b$

$f(-1) = -a$ ...(2)

$\therefore$ from (1) and (2)

$$-a = \frac{1}{9}-\frac{2}{3}\ell n\frac{2}{3}$$

$$a = \frac{2}{3}\ell n\left(\frac{2}{3}\right)-\frac{1}{9}$$

$\therefore\ g(x) = ax$

$$g(3) = 3\left(\frac{2}{3}\ell n\left(\frac{2}{3}\right)-\frac{1}{9}\right)$$

$$g(3) = 2\ell n\left(\frac{2}{3}\right)-\frac{1}{3}$$

$$g(3) = \ell n\frac{4}{9}-\frac{1}{3}$$

$$g(3) = \ell n\frac{4}{9}-\ell n e^{\frac{1}{3}}$$

$$g(3) = \ell n\left(\frac{4}{9\cdot e^{\frac{1}{3}}}\right)$$

$$g(3) = \log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$$

15. Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in R$. Then $f'(10)$ is equal to _____.

[JEE(Main)Jan.2024]

माना $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in R$ है। तो $f'(10)$ बराबर है _____।

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Ans. 202

Sol. $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$

$f'(x) = 3x^2 + 2xf'(1) + f''(2)$

$f''(x) = 6x + 2f'(1)$

$f'''(x) = 6$

$\Rightarrow f'''(3) = 6$

$f''(2) = 12 + 2f'(1)$ _____(1)

$f'(1) = 3 + 2f'(1) + f''(2)$ _____(2)

adding (1) and (2)

$0 = 15 + 3f'(x) \Rightarrow f'(1) = -5$

$f''(2) = 12 - 10 = 2$

Now

$f'(10) = 300 + 20(-5) + 2$

$= 300 - 100 + 2 = 202$

16. If $y = \frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$, then $96\, y'\left(\frac{\pi}{6}\right)$ is equal to _____.

[JEE(Main)Jan.2024]

यदि $y = \frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$ है, तो $96\, y'\left(\frac{\pi}{6}\right)$ बराबर है _____

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Ans. 105

Sol. $y = \frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(x\sqrt{x}-1\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2\cdot x - 5\right)\cos^3 x$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(\sqrt{x}\right)^3-1^3\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left[3\cos^5 x - 5\cos^3 x\right]$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(\sqrt{x}-1\right)\left(x+\sqrt{x}+1\right)}{\sqrt{x}(x+\sqrt{x}+1)} + \frac{1}{15}\left[3\cos^5 x - 5\cos^3 x\right]$

$y = (x-1) + \frac{1}{15}\left[3\cos^5 x - 5\cos^3 x\right]$

Now, diff w.r.t x

$y' = 1 - 0 + \frac{1}{15}\left[15\cos^4 x\cdot(-\sin x) - 15\cos^2 x(-\sin x)\right]$

$$y' = 1 - \sin x \cdot \left[\cos^4 x - \cos^2 x\right]$$

$$y'\left(\frac{\pi}{6}\right) = 1 - \sin\frac{\pi}{6}\left[\cos^4\frac{\pi}{6} - \cos^2\frac{\pi}{6}\right]$$

$$= 1 - \frac{1}{2}\left[\frac{9}{16} - \frac{3}{4}\right]$$

$$= 1 - \frac{1}{2}\left[\frac{9-12}{16}\right] \Rightarrow 1 + \frac{3}{32} \Rightarrow \frac{35}{32}$$

$$96.y'\left(\frac{\pi}{6}\right) = 96 \times \frac{35}{32} \Rightarrow 105$$

17. Let $f(x) = x^5 + 2e^{\frac{x}{4}}$ for all $x \in R$. Consider a function g(x) such that (gof) (x) = x for all $x \in R$. Then the value of $8g'(2)$ is : **[JEE(Main)April,2024]**

माना सभी $x \in R$ के लिए $f(x) = x^5 + 2e^{\frac{x}{4}}$ है। एक फलन g(x), जिसके लिए (gof) (x) = x $\forall$ $x \in R$ है, का विचार कीजिए। तो $8g'(2)$ का मान है : **[JEE(Main)April,2024]**

(A) 8 (B) 4 (C) 2 (D) 16

Ans. D

Sol. g(f(n)) = x

Differentiate w.r.t x, we get

$$g'(f(x)) \cdot f'(x) = 1$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

We need to find g'(2), so,

$f(0) = 2 \Rightarrow f(x) = x^5 + 2e^{x/4}$

$$f'(x) = 5x^4 + \frac{1}{2}e^{\frac{x}{4}}$$

$$f'(x) = \frac{1}{2}$$

$$\therefore\ g'(f(0)) = \frac{1}{f'(0)}$$

$$g'(2) = 2$$

$\therefore 8g'(2)=16$

18. Let $f(x)=x^5+2x^3+3x+1$, $x \in \mathbf{R}$, and $g(x)$ be a function such that $g(f(x))=x$ for all $x \in \mathbf{R}$. the $\frac{g(7)}{g'(7)}$ is equal to : **[JEE(Main)April,2024]**

माना $f(x)=x^5+2x^3+3x+1$, $x \in \mathbf{R}$ है, तथा एक फलन $g(x)$ इस प्रकार है कि $g(f(x))=x \ \forall \ x \in R$ है। तो $\frac{g(7)}{g'(7)}$ बराबर है: **[JEE(Main)April,2024]**

(A) 42 (B) 7 (C) 14 (D) 1

Ans. C

Sol. $f(x)=x^5+2x^3+3x+1$

$f'(x)=5x^4+6x^2+3$

$f'(1)=5+6+3=14$

$g(f(x))=x$

$g'(f(x))f'(x)=1$

for $f(x)=7$

$\Rightarrow x^5+2x^3+3x+1=7$

$\Rightarrow x=1$

$g'(7)\ f'(1)=1 \Rightarrow g'(7)=\frac{1}{f'(1)}=\frac{1}{14}$

$x=1, f(x)=7 \Rightarrow g(7)=1$

$\frac{g(7)}{g'(7)}=\frac{1}{1/14}=14$

19. If $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$, then at $\theta=\frac{\pi}{2}$, $y''+y'+y$ is equal to : **[JEE(Main)April,2024]**

यदि $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$ है, तो $\theta=\frac{\pi}{2}$, $y''+y'+y$ बराबर है : **[JEE(Main)April,2024]**

(A) 2 (B) 1/2 (C) 1 (D) 3/2

Ans. A

Sol. $y=\dfrac{2\cos\theta+2\cos^2\theta-1}{4\cos^3\theta-3\cos\theta+8\cos^2\theta-4+5\cos\theta+2}$

$$y=\frac{\left(2\cos^2\theta+2\cos\theta-1\right)}{\left(2\cos^2\theta+2\cos\theta-1\right)\left(2\cos\theta+2\right)}$$

$$y=\frac{1}{2}\left(\frac{1}{1+\cos\theta}\right)$$

$$\Rightarrow \theta=\frac{\pi}{2} \qquad y=\frac{1}{2}$$

$$y'=\frac{1}{2}\left(\frac{-1}{\left(1+\cos\theta\right)^2}\times\left(-\sin\theta\right)\right)$$

$$\Rightarrow \theta=\frac{\pi}{2} \qquad y'=\frac{1}{2}$$

$$y''=\frac{1}{2}\left[\frac{\cos\theta\left(1+\cos\theta\right)^2-\sin\theta\left(2\right)\left(1+\cos\theta\right)\left(-\sin\theta\right)}{\left(1+\cos\theta\right)^4}\right]$$

$$\Rightarrow \theta=\frac{\pi}{2} \qquad y''=1$$

20. Let $f:(-\infty,\infty)-\{0\}\to\mathbb{R}$ be a differentiable function such that $f'(1)=\lim\limits_{a\to\infty}a^2f\left(\frac{1}{a}\right)$. Then $\lim\limits_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ is equal to : **[JEE(Main)April,2024]**

माना $f:(-\infty,\infty)-\{0\}\to\mathbb{R}$ एक अवकलनीय फलन इस प्रकार का है कि $f'(1)=\lim\limits_{a\to\infty}a^2f\left(\frac{1}{a}\right)$, तो $\lim\limits_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{5}{2}+\frac{\pi}{8}$ (B) $\frac{3}{4}+\frac{\pi}{8}$ (C) $\frac{3}{8}+\frac{\pi}{4}$ (D) $\frac{3}{2}+\frac{\pi}{4}$

Ans. A

Sol. $f'(1)=\lim\limits_{a\to\infty}a^2f\left(\frac{1}{a}\right)$

So $f\left(\frac{1}{a}\right)=0$ should be at $x\to\infty$

Now $\lim\limits_{a\to\infty} a^2\left[\left[\frac{1}{2}+\frac{1}{2a}\right]\tan^{-1}\left(\frac{1}{a}\right)+1+\frac{\ell n a^{-2}}{a^2}\right]$

$f\left(\frac{1}{a}\right)=\left[\frac{1}{2}+\frac{1}{2a}\right]\tan^{-1}\left(\frac{1}{a}\right)+1+\frac{\ell n\left(\frac{1}{a^2}\right)}{a^2}$ (according to question $f\left(\frac{1}{a}\right)$ should be zero whenever a tending to infinite but it is 1 so there is a conceptual mistake in the question)

$f(x)=\left(\frac{1}{2}+\frac{x}{2}\right)\tan^{-1}x+1+x^2\ell n\left(x^2\right)$

$f'(x)=\frac{x+1}{2}\cdot\frac{1}{1+x^2}+\frac{\tan^{-1}x}{2}+\frac{x^2}{x^2}2x+\ell nx^2\cdot 2x$

According to NTA $f'(1)=1\times\frac{1}{2}+\frac{\pi}{8}+2=\frac{5}{2}+\frac{\pi}{8}$

21. If $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & X\neq 0\\ 0, & X=0\end{cases}$ then: **[JEE(Main)April,2024]**

यदि $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & X\neq 0\\ 0, & X=0\end{cases}$ तो : **[JEE(Main)April,2024]**

(A) $f''\left(\frac{2}{\pi}\right)=\frac{24-\pi^2}{2\pi}$ (B) $f''\left(\frac{2}{\pi}\right)=\frac{12-\pi^2}{2\pi}$ (C) $f''(0)=1$ (D) $f''(0)=0$

Ans. A

Sol. $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & X\neq 0\\ 0, & X=0\end{cases}$

$\lim\limits_{h\to 0^+} f(0+h)=\lim\limits_{h\to 0}(0+h)^3\cdot\sin\left(\frac{1}{0+h}\right)$

$\Rightarrow \lim\limits_{h\to 0^+} h^3\cdot\sin\left(\frac{1}{h}\right)$

$\Rightarrow 0\times(-1,1]=0$

$\lim\limits_{h\to 0^-} f(0-h)=\lim\limits_{h\to 0}(0-h)^3\sin\left(-\frac{1}{h}\right)$

$\Rightarrow 0 \times [-1,1] = 0$

f(x) is continuous

$$f'(x) = 3 \cdot x^2 \cdot \sin\left(\frac{1}{x}\right) - x \cdot \cos\frac{1}{x}$$

$$f''(x) = 6x \cdot \sin\frac{1}{x} - 3\cos\frac{1}{x} - \cos\left(\frac{1}{x}\right) - \frac{\sin\frac{1}{x}}{x}$$

$$f''\left(\frac{2}{\pi}\right) = \frac{24-\pi^2}{2\pi}$$

22. Suppose for a differentiable function h, h(0)=0, h(1)=1 and $h'(0) = h'(1) = 2$. If $g(x) = h\left(e^x\right)e^{h(x)}$, then $g'(0)$ is equal to : **[JEE(Main)April,2024]**

मान लीजिए कि अवकलनीय फलन h के लिए, h(0)=0, h(1)=1 तथा $h'(0) = h'(1) = 2$ हैं। यदि $g(x) = h\left(e^x\right)e^{h(x)}$, तो $g'(0)$ बराबर है : **[JEE(Main)April,2024]**

(A) 3 (B) 8 (C) 4 (D) 5

Ans. C

Sol. $g(x) = h(e^x).e^{h(x)}$

$g'(x) = h(e^x).e^{h(x)}.h'(x) + e^{h(x)}h'(e^x).e^x$

$g'(0) = h(1)^{eh(0)}h'(0) + e^{h(0)}h'(1)$

$= 2 + 2 = 4$

23. Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to : **[JEE(Main)April,2024]**

माना $f(x) = ax^3 + bx^2 + cx + 41$ इस प्रकार है कि $f(1) = 40$, $f'(1) = 2$ तथा $f''(1) = 4$ है। तो $a^2 + b^2 + c^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 73 (B) 62 (C) 51 (D) 54

Ans. C

Sol. $f(x) = ax^3 + bx^2 + cx + 41$

$f'(x) = 3ax^2 + 2bx + c$

$\Rightarrow f'(1) = 3a + 2b + c = 2$ .....(1)

$f''(n) = 6ax + 2b$

$\Rightarrow f''(1) = 6a + 2b = 4$

$3a + b = 2$ .....(2)

(A) – (2) $\Rightarrow$ b + c = 0 ...(3)

f(1) = 40

a + b + c + 41 = 40

use (3)

a + 41 = 40

by (2)

– 3 + b = 2 $\Rightarrow$ b = 5 & c = – 5

$a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$

24. If $\log_e y = 3\sin^{-1}x$, then $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ is equal to : **[JEE(Main)April,2024]**

यदि $\log_e y = 3\sin^{-1}x$ है, तो $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ बराबर है : **[JEE(Main)April,2024]**

(A) $9e^{\frac{\pi}{6}}$ (B) $3e^{\frac{\pi}{2}}$ (C) $3e^{\frac{\pi}{6}}$ (D) $9e^{\frac{\pi}{2}}$

Ans. D

Sol. $\log_e y = 3\sin^{-1}x$

$y = e^{3\sin^{-1}x}$

$\frac{dy}{dx} = e^{3\sin^{-1}x} \cdot \frac{3}{\sqrt{1-x^2}}$

$\sqrt{1-x^2}\frac{dy}{dx} = 3y$

Again differentiate

$\sqrt{1-x^2} \cdot y'' - \frac{2x}{2\sqrt{1-x^2}} y' = 3y'$

$(1-x)^2 y'' - xy' = 3y'\left(\sqrt{1-x^2}\right)$

So value of $3y'\left(\sqrt{1-x^2}\right)$

at $x = \frac{1}{2}$

$3 \cdot \frac{3}{\sqrt{1-x^2}} e^{\sin^{-1}x}\left(\sqrt{1-x^2}\right)$

$= 9e^{3\frac{\pi}{6}} = 9e^{\frac{\pi}{2}}$

25. The value of $\lim\limits_{x\to 0} 2\left(\dfrac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____. **[JEE(Main)April,2024]**

$\lim\limits_{x\to 0} 2\left(\dfrac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ का मान होगा _____ **[JEE(Main)April,2024]**

Ans. 55

Sol. $\lim\limits_{x\to 0}\dfrac{2}{x^2}\left(1-\left(1-\dfrac{x^2}{2}\right)\left(1-\dfrac{2^2x^2}{2}\right)^{\frac{1}{2}}\left(1-\dfrac{3^2x^2}{2}\right)^{\frac{1}{3}}.....\left(1-\dfrac{10^2x^2}{2}\right)^{\frac{1}{10}}\right)$

$\lim\limits_{x\to 0}\dfrac{2}{x^2}\left(1-\left(1-\dfrac{x^2}{2}\right)\left(1-\dfrac{2^2x^2}{2.2}\right)\left(1-\dfrac{3^2\cdot x^2}{3.2}\right).....\left(1-\dfrac{10^2x^2}{2.10}\right)\right)$

$= \lim\limits_{x\to 0}\dfrac{2}{x^2}\left(1-\left(1-\dfrac{x^2}{2}-\dfrac{2x^2}{2}-\dfrac{3x^2}{2}.....-\dfrac{10x^2}{2}\right)\right)$

$= \left(\dfrac{2}{2}(1+2+3+.....+10)\right) = \dfrac{10(10+1)}{2} = 55$